MIL-R-43825B
25 July 1978
SUPERSEDING
MIL-R-0043825A (GL)
25 July 1977, and
MIL-R-43825
22 November 1972

MILITARY SPECIFICATION

RACKS, TRAY BUSSING, MOBILE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two-compartment mobile racks for bussing trays of soiled dishes.

1.2 Classification. Racks covered by this specification shall be of the following types and sizes as specified (see 6.2).

Type I - To carry 14 by 18 inch (356 by 475 mm) rectangular trays
  Size 1 Capacity 22 trays
  Size 2 Capacity 44 trays

Type II - To carry 15 by 20 inch (381 by 508 mm) rectangular trays
  Size 1 Capacity 22 trays
  Size 2 Capacity 44 trays

Type III - To carry 14 by 18 inch (356 by 457 mm) trapezoidal trays, capacity 44 trays

Type IV - To carry 10-3/4 by 13-7/8 inch (273 by 352 mm) rectangular trays, capacity 48 trays

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Natick Research and Development Command, ATTN: DRDNA-ES, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7320

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:29Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-T-776 - Tray, Service (Laminated Fiber Glass, Polyester Resin).
FF-C-77 - Casters, Institutional Duty.

MILITARY

MIL-B-2316 - Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery of.
MIL-T-43259 - Trays, Service, Plastic (Laminated Fiber Glass Reinforced Polyester Resin).
MIL-T-43271 - Tray, Mess, Compartmented, Plastic.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL SANITATION FOUNDATION (NSF)

Standard 2 - Food Service Equipment

Listing of Food Service Equipment

(Application for copies should be addressed to the National Sanitation Foundation, NSF Building, Ann Arbor, MI 48105.)

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:29Z
Check the source to verify that this is the current version before use.

U.S. DEPARTMENT OF COMMERCE

CS 245-62 - Vinyl-Metal Laminates

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The rack shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modification to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. Racks shall comply with the applicable requirements of NSF Standard 2.

3.2.1 Compliance. Prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the racks he proposes to furnish under this specification meet the applicable requirements of NSF as follows:

3.2.1.1 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished rack, or

(2) A certification for the rack issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the racks are being procured, indicating that the racks have been tested and conform to NSF Standard 2.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:29Z
Check the source to verify that this is the current version before use.

3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial racks provided the completed item complies with all provisions of this specification (see 6.4).

3.3.1 Vinyl-metal laminates. Vinyl-metal laminates used on the racks shall conform to CS 245.

3.4 Design and construction. All racks shall be of double compartment design, consisting of an aluminum alloy frame, extruded aluminum alloy tray slides, two aluminum side panels with decorative exterior coating, a closed bottom, and four swivel casters mounted under the frame. The casters shall be attached to a reinforcing channel or reinforcing plate the entire depth of the rack and shall be of sufficient width to fully receive the plate area of the casters. Type I racks shall be designed for the 14 by 18 by 1 inch (nominal) (356 by 457 by 25.4 mm) trays designated as type I, size 2 of L-T-776 and size 1 of MIL-T-43259. Type II racks shall be designed for the 15 by 20 by 1 inch (nominal) (381 by 508 by 25.4 mm) trays designated as type I, size 1 of L-T-776 and size 2 of MIL-T-43259. Type III racks shall be designed for the 14 by 18 by 1 inch (nominal) (356 by 457 by 25.4 mm) trapezoidal trays designated as type II of L-T-776. Type IV racks shall be designed for the 10-3/4 by 13-7/8 by 1-1/8 inch (273 by 352 by 28.6 mm) rectangular compartmented tray conforming to MIL-T-43271. The base plate shall be an integral part of the rack, but the side panels may be removable. The frame shall be contructed perpendicular to the base and the top of the frame shall be parallel to the base. The rack shall accept the applicable size tray without binding, and when fully loaded with trays, no part of any tray shall extend beyond the planes established by the outside surface of the corner posts. The racks shall show no evidence of breakage, looseness of parts, parts separation, unevenness of castors with rack standing unloaded on a level surface, or permanent deformation in excess of 3/16 inch (4.8 mm) when tested as specified in 4.4.1, 4.4.2 and 4.4.3.

3.4.1 Capacity and dimensions. Overall dimensions for the racks shall not include the bumpers, and shall have a maximum allowable tolerance of ± 2-1/2 inches (63.5 mm).

3.4.1.1 Type I, size 1. Type I, size 1 racks shall carry the specified trays end loaded, one tray per pair of tray slides. The minimum capacity shall be twenty-two trays. Overall dimensions shall be: width 36 inches (914 mm), depth 19 inches (483 mm), and height 70 inches (1778 mm).

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:29Z
Check the source to verify that this is the current version before use.

3.4.1.2 Type I, size 2 and type III. Type I, size 2 and type III racks shall carry the specified trays side loaded, two trays per pair of tray slides. The minimum capacity shall be forty-four trays. Overall dimensions shall be: width 42 inches (1067 mm), depth 31 inches (787 mm), and height 70 inches (1778 mm).

3.4.1.3 Type II, size 1. Type II, size 1 racks shall carry the specified trays end loaded, one tray per pair of tray slides. The minimum capacity shall be twenty-two trays. Overall dimensions shall be: width 36 inches (914 mm), depth 24 inches (610 mm), and height 70 inches (1778 mm).

3.4.1.4 Type II, size 2. Type II, size 2 racks shall carry the specified trays side loaded, two trays per pair of tray slides. The minimum capacity shall be forty-four trays. Overall dimensions shall be: width 46 inches (1168 mm), depth 31 inches (787 mm), and height 70 inches (1778 mm).

3.4.1.5 Type IV. The type IV racks shall carry the specified trays side loaded, two trays per pair of tray slides. The minimum capacity shall be forty-eight trays. Overall dimensions shall be: width 33 inches (838 mm), depth 22 inches (559 mm), height 70 inches (1778 mm).

3.4.2 Side panels. The side panels of the rack shall have a decorative exterior coating. The coating shall be a vinyl laminate conforming to the requirements specified in 3.3.1. Unless otherwise specified (see 6.2), the coating shall be a walnut wood grain. Side panels are not required for type IV racks. Side panels shall be securely fastened at the bottom, top and sides by threaded fasteners and shall not rattle or vibrate when tested as specified in 4.4.1 and 4.4.3.

3.4.3 Casters. Each bussing rack shall be provided with four casters conforming to type II, class B, 5 inch (127 mm) diameter, style B or C, mounting P (Plate) of FF-C-77. The two casters on the side to which the push handle is attached shall be furnished with locking brakes. Casters shall be securely fastened to reinforcing channels or reinforcing plate with 1/4 inch minimum diameter bolts, lockwashers and nuts. The casters shall show no evidence of damage when tested as specified in 4.4.1, 4.4.2 and 4.4.3.

3.4.4 Push handle. A push handle shall be attached to one side of all racks. The handle shall be a rigid metal tube of no less than 1 inch (25.4 mm) diameter or 3/4 inch (19 mm) diameter solid rod and shall have no less than 1-1/2 inches (38 mm) space between the handle and the side panel of the rack. If aluminum is used, it shall be anodized or polyvinyl coated. The handle shall be located 42 ± 2 inches (1067 ± 351 mm) above the floor, and shall be protected by non-marking rubber bumpers on the handle.

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:29Z
Check the source to verify that this is the current version before use.

3.4.5 Tray slides. Type I, II and IV racks shall have extruded aluminum angle tray slides. Type III racks shall have extruded aluminum channel or wide profile angle tray slides designed to keep loaded trapezoidal trays from tipping. Type I, II and III racks shall have tray slides mounted with a spacing of 5-5/8 ± 1/8 inch (143 ± 3 mm). Type IV racks shall have tray slides mounted with a spacing of 4-3/4 ± 1/4 inch (121 ± 6 mm). No tray slide shall be damaged and no tray shall fall through the tray slides when tested as specified in 4.4.1, 4.4.2 and 4.4.3.

3.4.6 Bumpers. All racks shall have replaceable, non-marking rubber bumpers attached to the lower end of each corner, or doughnut type bumper, or a replaceable non-marking rubber perimeter bumper attached completely around the perimeter of the base of the rack. The push handle shall be protected by non-marking rubber bumpers on the handle as specified in 3.4.4. Vertical bumpers shall wrap around the two surfaces of each corner. The bumpers shall be applied so that no part of the rack touches the wall except the bumper when tested in 4.4.1.

3.5 Finish. Aluminum alloy finish shall be the manufacturer's standard finish provided the finish meets the NSF requirements.

3.6 Marking for identification. Each rack shall be permanently and legibly marked with the manufacturer's name or trademark readily identifiable with said manufacturer, national stock number and contract number.

3.7 Workmanship. The finished bussing racks shall be clean and free of scratches, dents, breaks, sharp edges, sharp corners, deformities, burrs and slivers. Rivets shall be tight and the rivet heads and peened ends shall be round and concentric and shall be in full contact with the joined metal surfaces. Use of hollow rivets and self-expanding pop rivets are not acceptable. Fusion welds shall be clean, continuous, free of pits, scale, slag and splatter. Threaded fasteners shall be drawn tight. Lockwashers shall be used on all bolts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:29Z
Check the source to verify that this is the current version before use.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2.1 Visual examination. The rack shall be examined for defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5, expressed in terms of defects per hundred units.

TABLE I. Classification of visual defects

Examine	Defect
Standard product	Item not in accordance with manufacturer's standard product except for changes necessary to comply with specification requirements
Finish	Dented or deformed
	Scratched
	Sharp corners or edges, burrs or slivers
	Decorative coating on side panels not color and texture specified
	Decorative coating on side panels separated from metal, peeled or torn
Design and construction	Not type and size specified
	Rubber bumpers missing or not as specified
	Rubber bumpers loosely attached
	Handle missing on size 2 of types I and II, and type III racks, not as specified
	Push handle bumpers missing
	On flat surface all four wheels not touching surface
	Side panels not provided

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:29Z
Check the source to verify that this is the current version before use.

TABLE I. Classification of visual defects (cont'd)

Examine	Defect
Design and construction (cont'd)	Not enclosed on 2 sides and bottom Base plate not integral part of rack Side panels not secured along entire bottom edge, top edge and vertical edges Casters not attached to reinforcing channel or reinforcing plate not provided
Rivets	Missing, loose, not fully preened, not in full contact with joined members Hollow type or self-expanding pop rivets used in construction
Welding	Any weld not continuous, not sound, or cracked Any weld containing scale, slag, pits, or splatter
Threaded fasteners	Missing, stripped or fractured Loose Lockwashers not used with bolts
Marking for identification	Missing, incomplete, illegible

4.2.2.2 Dimensional examination. Examination shall be made of the completed bussing racks to determine compliance with dimensions specified. Any dimensions not within specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in defects per hundred units.

4.2.2.3 End item testing. One representative production unit shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3. Failure of any test shall be cause for rejection of the lot.

4.2.3 Packaging inspection. Preservation-packaging, packing and marking shall be inspected in accordance with the quality assurance provisions of MIL-E-2316.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:29Z
Check the source to verify that this is the current version before use.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the specification requirements shall be examined and validated.

4.4 Test methods. In the following tests the sides of the rack are defined to be adjacent to the loading side.

4.4.1 Impact load. The rack shall be loaded to its full capacity with the specified trays (see 3.4). Each tray shall contain a distributed load so the total weight of the tray and load is not less than 7-1/2 pounds (3.4 kg). The rack shall be propelled into a hard, rigid wall at a speed not less than 30 feet (9.1 m) per 10 seconds in the following manner. The side opposite the rack handle shall impact the wall head-on so that both corners strike the wall simultaneously. This test shall be performed five times. Each corner opposite the rack handle shall impact the wall with the rack approaching the wall at a 30 degree angle. This test shall be performed five times. The rack shall then be unloaded and the above test performed on the handle end of the rack. The rack, rubber bumpers and the wall shall be inspected to determine compliance with 3.4, 3.4.2, 3.4.3, 3.4.5 and 3.4.6. Any nonconformance shall constitute failure of this test.

4.4.2 Shock load. The rack shall be loaded as in 4.4.1. The loaded rack shall be placed on a raised platform having vertical faces not less than 2 inches (51 mm) in height. The loaded rack shall be rolled completely off the platform onto a level concrete surface at a speed not less than 30 feet (9.1 m) per 10 seconds. This test drop shall be repeated five times each side, with the rack positioned so the side is parallel to the step, and with the direction of rack travel perpendicular to the step. The rack shall be examined to determine conformance with 3.4, 3.4.3 and 3.4.5. Any nonconformance shall constitute failure of this test.

4.4.3 Obstacle test. The rack shall be loaded as specified in 4.4.1. The loaded rack shall pass over a 1/4 inch (6.4 mm) obstacle ten times at a minimum speed of 30 feet (9.1 m) per 10 seconds to determine compliance with 3.4, 3.4.2, 3.4.3 and 3.4.5. Any nonconformance shall constitute failure of this test.

5. PACKAGING

5.1 Levels A, B and C. Preservation-packaging, packing and marking for each level of protection shall be in accordance with MIL-B-2316 (see 6.2).

6. NOTES

6.1 Intended use. The racks covered by this specification are intended for temporary storage of trays containing soiled dishes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:29Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The type and size required (see 1.2).
(c) Color and texture of vinyl laminate when other than walnut wood grain is required (see 3.4.2 and 6.3).
(d) Selection of the applicable levels of preservation-packaging, packing and marking (see 5.1).

6.3 Decorative coating. Colors and textures other than walnut wood grain, when required, shall conform to the manufacturer's standard colors and textures.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

6.5 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM A 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

Custodians:

Army - GL
Navy - SA
Air Force - 99

Review activities:

Army - MD
Navy - MC

Preparing activity:

Army - GL

Project No. 7320-0669

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:29Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
US ARMY NATICK RESEARCH and DEVELOPMENT COMMAND
NATICK, MASSACHUSETTS 01760

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
DRDNA-LS

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY
DOD – 314

Commander
US Army Natick Research and Development Command
ATTN: DRDNA-LS
Natick, MA 01760

NATICK OP 1 Apr 76
FOLD

Source: https://assist.dla.mil -- Downloaded: 202610-21T22:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-R-43825L -- Racks, Tray Bussing, Mobile

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T22:29Z
Check the source to verify that this is the current version before use.